

PETU Strong Predictions and Preregistered Tests

Formal prediction classes, null models, thresholds and domain templates

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Claim status and scope

This document is a preregistration-oriented companion to the PETU mathematical formalism and empirical atlas. It converts triadic structural closure into explicit empirical predictions with frozen definitions, baseline models, null models, thresholds and failure conditions. It is not a claim that the framework has already become a universal law.

Public records. Preprint DOI: <https://doi.org/10.5281/zenodo.20821557>. Data/code DOI: <https://doi.org/10.5281/zenodo.20820252>. Repository: <https://github.com/hector-eag85/petu-triadic-structural-closure-atlas>.

Abstract

This document defines strong predictions and preregistered test families for triadic structural closure. The framework predicts that coherent natural systems can often be represented by the integrated relation of support (H), structure (S) and linkage (L), with system-level closure denoted by Φ . The document specifies prediction classes, model hierarchies, null models, baseline comparisons, threshold rules, reporting metrics, falsification criteria and domain-specific preregistration templates. It distinguishes internal closure reconstruction from external endpoint prediction and requires separation of raw support H_{raw} from structured support $H_{\text{structured}}$ wherever support may already contain structural information.

1. General preregistration principle

For each domain X , the following must be specified before analysis:

1. dataset, version, access date and inclusion/exclusion rules;
2. natural observational units Ω_X ;
3. perturbative or null units;
4. definitions of H, S, L, Φ ;
5. whether $H = H_{\text{raw}}$ or $H = H_{\text{structured}}$;
6. primary endpoint and secondary endpoints;
7. monadic, dyadic and triadic baselines;
8. null models;
9. validation strategy;
10. statistical thresholds;
11. failure criteria;
12. interpretation rules.

No domain should be counted as strong confirmatory evidence unless these definitions are frozen before analysis.

2. Required model hierarchy

Every preregistered test should evaluate:

$$\begin{aligned} M_H : \Phi \sim H, \quad M_S : \Phi \sim S, \quad M_L : \Phi \sim L, \\ M_{HS} : \Phi \sim H + S, \quad M_{HL} : \Phi \sim H + L, \quad M_{SL} : \Phi \sim S + L, \\ M_{HSL} : \Phi \sim H + S + L, \end{aligned}$$

and, where justified,

$$M_{int} : \Phi \sim H + S + L + HS + HL + SL + HSL.$$

The triadic claim is strongest when M_{HSL} or M_{int} outperforms monadic and dyadic baselines while remaining interpretable and resistant to null models.

3. Performance metrics

Each test should declare its primary performance functional P . Acceptable metrics include cross-validated R^2 , Spearman correlation, AUC, mean absolute error reduction, permutation p -values, null z -scores or domain-specific predictive loss. Every report should include:

- sample size and group structure;
- number of natural and perturbed/null observations;
- number of scales;
- H -only performance;
- best dyadic baseline;
- HSL performance;
- triadic gain G_3^{diff} ;
- null performance;
- evidence grade;
- failure modes.

4. Prediction classes

4.1. P1. Coherence reconstruction

Prediction:

$$\Phi_X \approx \mathcal{F}_X(H_X, S_X, L_X).$$

Primary test:

$$\Phi_X \sim H_X + S_X + L_X.$$

Initial success threshold:

$$R_{CV}^2 > 0.25 \quad \text{or} \quad \rho_{\text{Spearman}} > 0.50.$$

Failure occurs if $P(HSL \rightarrow \Phi) \leq \tau_\Phi$ under frozen definitions.

4.2. P2. Unity reconstruction

When an external unity endpoint $U_X^{external}$ exists:

$$U_X^{external} \approx \mathcal{U}_X(H_X, S_X, L_X).$$

If no external endpoint exists, report

$$U_X = (H_X S_X L_X)^{1/3}$$

as a derived PETU index, not as an independent endpoint.

4.3. P3. Irreducible triadic gain

Define

$$\mathcal{B} = \{H, S, L, H + S, H + L, S + L\}.$$

Prediction:

$$G_3^{diff} = P(H + S + L \rightarrow \Phi) - \max_{\mathcal{B}} P(\mathcal{B}) > \tau_G.$$

Recommended initial thresholds:

$$\Delta R_{CV}^2 > 0.05, \quad \Delta AUC > 0.03, \quad \Delta MAE > 5\%.$$

4.4. P4. Linkage emergence from structured support

The clean emergence prediction is

$$P(H_{\text{raw}} + S \rightarrow L) - P(H_{\text{raw}} \rightarrow L) > \tau_L.$$

If only $H_{\text{structured}}$ is available, apparent $H \rightarrow L$ must be labelled as structured-support prediction, not pure support prediction.

4.5. P5. Perturbative coherence loss

For a relevant disruptive perturbation π_i :

$$\Phi_X > \Phi_{\pi_i(X)} \quad \text{or} \quad F_{\pi_i(X)} > F_X.$$

Define

$$\Delta_{\Phi}^i = \text{median}(\Phi_X) - \text{median}(\Phi_{\pi_i(X)}).$$

Recommended threshold:

$$\Delta_{\Phi}^i > 0.05 \quad \text{or} \quad \text{relative coherence loss} > 5\%.$$

4.6. P6. Fragility from disproportion

Prediction:

$$F_X = \alpha(1 - U_X) + \beta(1 - B_X) + \gamma D_X + \delta R_X.$$

Primary test:

$$F_X \sim 1 - U_X + 1 - B_X + D_X + R_X.$$

Initial success threshold:

$$R_{CV}^2 > 0.25 \quad \text{or} \quad AUC > 0.70$$

for binary stable/fragile classification.

4.7. P7. Natural-versus-perturbed discrimination

Let $Y = 1$ for natural observations and $Y = 0$ for perturbative/null observations. Prediction:

$$AUC(HSL) > AUC(H)$$

with

$$\Delta AUC > 0.03.$$

Stronger support is

$$AUC(HSL) > 0.75.$$

4.8. P8. Multi-scale stability

For scales $r \in \mathcal{R}_X$:

$$\Phi_X(r) \approx \mathcal{F}_{X,r}(H_X(r), S_X(r), L_X(r)).$$

Closure is scale-stable if

$$\frac{\#\{r : P_r > \tau_\Phi\}}{|\mathcal{R}_X|} \geq 0.70.$$

Failure occurs if closure appears only at a post hoc selected scale.

4.9. P9. External endpoint validity

If an independent endpoint Y_X exists:

$$Y_X \approx \mathcal{E}_X(H_X, S_X, L_X).$$

Success requires:

$$P(HSL \rightarrow Y) > P(H \rightarrow Y)$$

and $P(HSL \rightarrow Y) > \tau_Y$. Recommended thresholds:

$$AUC > 0.75 \quad \text{or} \quad \rho_{\text{Spearman}} > 0.40$$

for noisy external regression.

4.10. P10. Alternative operationalization robustness

Let $(H^{(a)}, S^{(a)}, L^{(a)})$ and $(H^{(b)}, S^{(b)}, L^{(b)})$ be two frozen operationalizations. The directional pattern should hold under at least two preregistered operationalizations. Failure occurs if the effect appears only under one post hoc feature choice.

4.11. P11. Adversarial null resistance

Nulls may include label shuffling, feature shuffling, within-group permutation, block shuffling, degree-preserving graph randomization, time-phase randomization, spatial nulls and matched-complexity random features. Success requires:

$$P_{\text{observed}} > P_{\text{null}} + \tau_N.$$

Recommended rule:

$$z_{\text{null}} > 2 \quad \text{or} \quad p_{\text{perm}} < 0.05,$$

with correction where applicable.

4.12. P12. Dynamic change prediction

For time-resolved systems:

$$\Delta\Phi_t \sim \Delta H_t + \Delta S_t + \Delta L_t.$$

Recommended threshold:

$$R_{CV}^2 > 0.20 \quad \text{or} \quad \rho_{\text{Spearman}} > 0.40.$$

5. Domain preregistration template

For each domain, record:

Field	Required content
Domain	Name and scale class: micro, meso or macro.
Dataset	Source, accession, version and access date.
Natural units	Definition of Ω_X .
H definition	Operational formula and whether $H = H_{\text{raw}}$ or $H = H_{\text{structured}}$.
S definition	Operational formula.
L definition	Operational formula.
Φ definition	Internal, external or derived endpoint; anti-circularity statement.
Primary prediction	Choose from P1–P12.
Baselines	$H, S, L, H + S, H + L, S + L, H + S + L$.
Nulls	Declared null models.
Cross-validation	Grouping variable and leakage controls.
Success criteria	Exact threshold values.
Failure criteria	Exact refutation or weakening conditions.

6. Domain-specific prediction map

Domain	Preregistered prediction focus
Microphysics	$HSL \rightarrow \Phi_{\text{micro}}$ beyond mass/energy-only baselines and matched-size nulls.
Proteins	Folding coherence from energetic support, structural compactness and residue-contact linkage; sequence and contact-map nulls.
Genomics	Genomic coherence from sequence support, organization and regulatory/compositional linkage; composition-preserving shuffles.
Metabolism	Network coherence from reaction support, topology and flux/connectivity linkage; structured-support exception expected.
Neurodynamics	State coherence from signal support, spectral/temporal structure and functional coupling; phase and channel nulls.
Cardiorespiratory	Physiological coherence from signal support, rhythm structure and cardiorespiratory coupling; subject-aware validation.
Embryogenesis	Developmental coherence from cellular support, transcriptional structure and lineage/signaling linkage.
Immune system	Immune-state coherence and external labels from expression support, cell-state structure and immune linkage.

Ecosystems	Resilience/stability from species/resource support, network structure and trophic linkage.
Hydrology	Regime coherence from flow support, temporal structure and linkage; clean $H_{\text{raw}} + S \rightarrow L$ test.
Climate	Regional coherence from thermal support, spatiotemporal structure and teleconnection linkage; block and phase nulls.
Cosmology	Cosmic-web coherence from density support, large-scale structure and filamentary linkage; density-matched nulls.

7. Cross-domain convergence test

Let $D = \{D_1, \dots, D_n\}$ be the set of domains and let P_d be the declared domain performance. Define

$$C_{\text{atlas}} = \frac{1}{n} \sum_{d=1}^n P_d.$$

Leave-one-domain-out convergence is

$$C_{\text{atlas}}^{(-d)} = \frac{1}{n-1} \sum_{j \neq d} P_j.$$

Cross-domain success requires

$$C_{\text{atlas}} > \tau_C \quad \text{and} \quad C_{\text{atlas}}^{(-d)} > \tau_C$$

for every domain d . The cross-domain claim fails if convergence depends entirely on one or two domains.

8. Evidence grading

Grade	Meaning
C	Internal $HSL \rightarrow \Phi$ positive, but limited sample and no perturbation or validation.
B	Internal reconstruction plus either perturbative contrast or classifier support.
A	Strong $HSL \rightarrow \Phi$ and $HSL \rightarrow U$, with null comparisons or cross-validation.
A+	Strong reconstruction, perturbative support and group-aware validation.
A++	Strong internal reconstruction, perturbative support, H-only controls, independent validation and external endpoint or robust cross-domain replication.

Evidence grades are descriptive labels and do not replace quantitative reporting.

9. Falsification matrix

The theory is weakened or falsified if:

- $HSL \rightarrow \Phi$ fails under frozen definitions;
- HSL does not beat H -only when H is genuinely raw;
- $H_{\text{raw}} + S$ does not improve linkage prediction;
- perturbations do not reduce Φ or increase F ;
- fragility is unrelated to unity, balance, dominance or rupture;
- external endpoints fail under anti-leakage validation;
- alternative operationalizations erase the effect;
- group-aware cross-validation fails;
- leave-one-domain-out convergence fails.

10. Minimal valid preregistered test

A minimal test requires:

1. one natural dataset;
2. frozen definitions of H, S, L, Φ ;
3. one H -only baseline;
4. one dyadic baseline;
5. one HSL model;
6. one null model;
7. one cross-validation strategy;
8. one success threshold;
9. one failure threshold;
10. a statement on whether H is raw or structured.

11. Strong preregistered test

A strong test includes all minimal requirements plus:

- group-aware validation;
- multiple null models;
- perturbative contrast;
- fragility prediction;
- external endpoint if available;
- alternative operationalization;
- leave-one-scale-out validation;
- public code and public derived outputs.

12. Master prediction set

$$P_1 : HSL \rightarrow \Phi$$

$$P_2 : HSL \rightarrow U$$

$$P_3 : G_3^{diff} > \tau_G$$

$$P_4 : H_{\text{raw}} + S \rightarrow L \quad > \quad H_{\text{raw}} \rightarrow L$$

$$P_5 : \pi(H, S, L) \rightarrow \downarrow \Phi$$

$$P_6 : (1 - U, 1 - B, D, R) \rightarrow F$$

$$P_7 : HSL \rightarrow Y_{\text{natural/perturbed}}$$

$$P_8 : HSL \rightarrow \Phi \text{ across scales}$$

$$P_9 : HSL \rightarrow Y_{\text{external}}$$

$$P_{10} : HSL > \text{adversarial nulls}$$

$$P_{11} : HSL \text{ robust under alternative operationalizations}$$

$$P_{12} : \Delta H, \Delta S, \Delta L \rightarrow \Delta \Phi$$

13. Final statement

PETU strong predictions convert triadic structural closure from a mathematical framework into a concrete empirical research program. The prediction set is designed to make the framework falsifiable through independent validation, adversarial nulls, alternative operationalizations and preregistered replication.